Lab 9: Investigating and Responding to Security Incidents

In this lab, Anti-Virus Guard (AVG) is used to perform a deep scan of an infected system, identify and quarantine affected files, and remove them from the system. All basic protection features are re-enabled. After researching the common malware types detected by the scan, we create a mock incident report detailing the cause, impact, containment, investigation, and recovery and repair of the system.

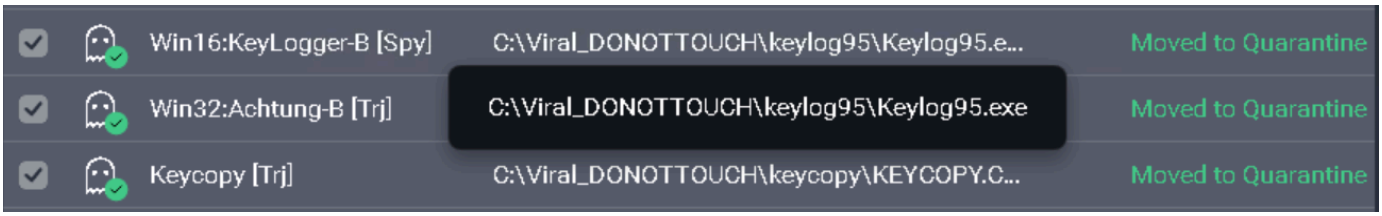
SECTION 1
Part 1: Prepare an Incident Response Report

Sample Incident Report: http://www.vlabsondemand.com/issa30/SecurityIncidentReport-Sample.pdf

Part 2: Use Antivirus Software to Scan the Infected System

Screenshot 1: Summary of the deep scan performed on TargetWindows02 with AVG

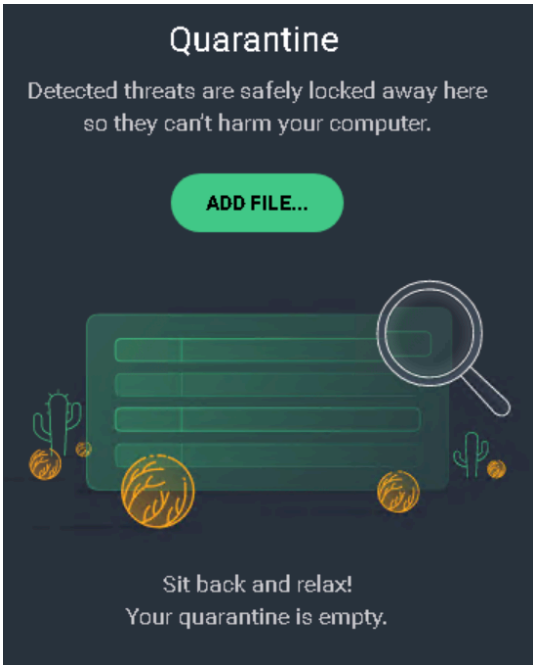
Part 3: Investigate and Remove the Identified Threats



Screenshot 2: Full file path of achtung.exe threat

Document any information you discover about eradicating this threat:

Achtung.exe is a worm that runs in the background of a device, recording keystrokes. Once keystrokes are logged, they are hidden in the machine for later retrieval or sent to the attacker. Personal information, possibly including username and password details, may be sent to a malicious actor. Most achtung.exe errors are the result of missing or corrupt versions of the executable file. The primary way to resolve these problems manually is to replace the EXE file with a fresh copy. It is highly recommended to clean up any invalid file path and EXE file extension references that could contribute to creating these achtung.exe error messages using antivirus software that will do a deep scan of the device. In addition, disconnecting the infected device from the network will prevent it from spreading to other devices. To eradicate this threat, quarantine and remove the infected files once the antivirus software identifies the worm and any infected files. To ensure the threat is mitigated, update the system and antivirus software, as worms often exploit vulnerabilities in outdated software. If the infection has caused significant damage or cannot be removed completely, restore the system from a backup taken before the infection occurred. Make sure the backup is clean and doesn't contain the worm. After ensuring the system is now malware-free, continue to monitor for signs of a possible recurrence. Sometimes, remnants of the worm may remain hidden or dormant and activate later.



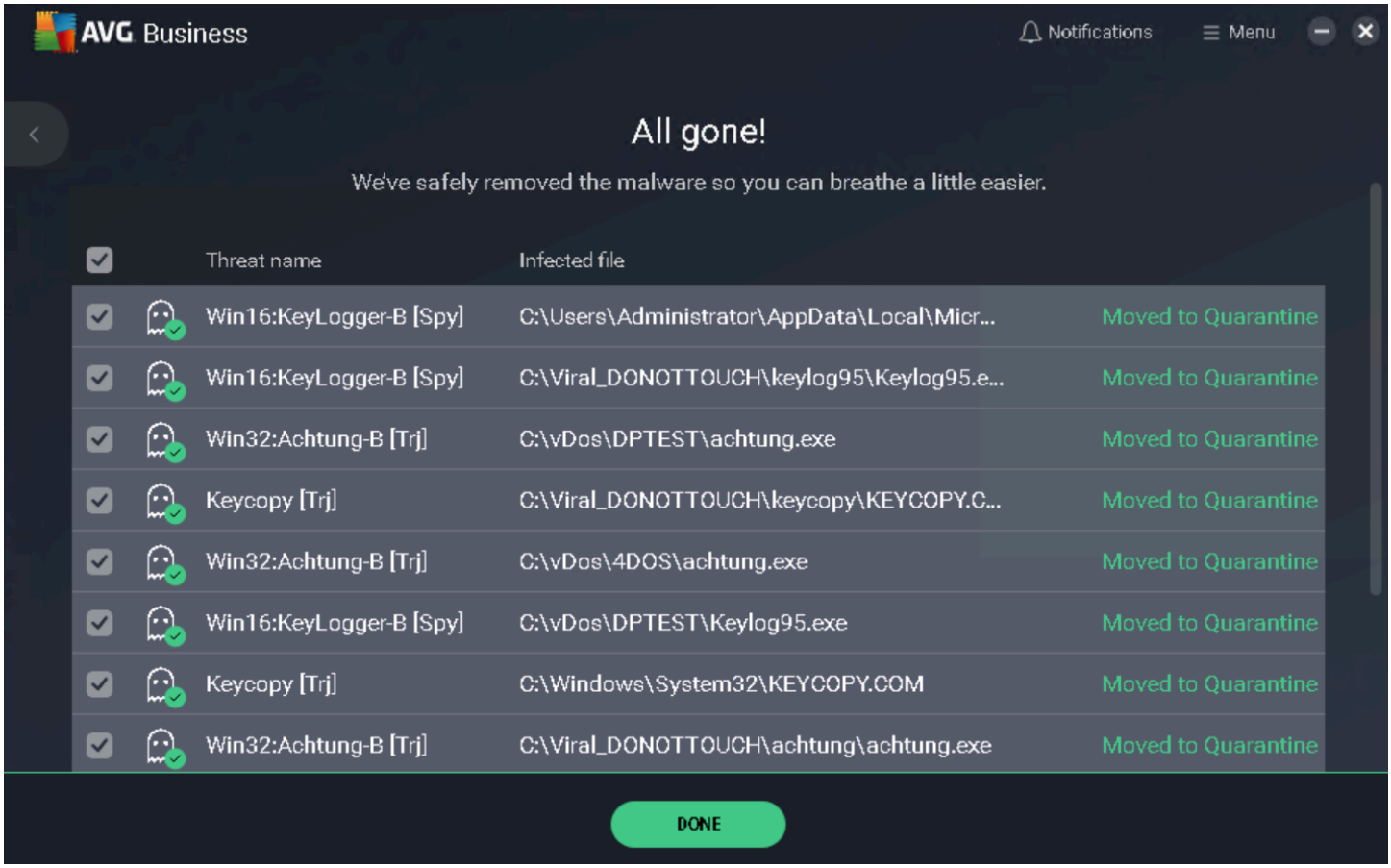
Screenshot 3: All infected files have been permanently deleted, leaving quarantine empty.

SECTION 2

Part 1: Prepare an Incident Response Report

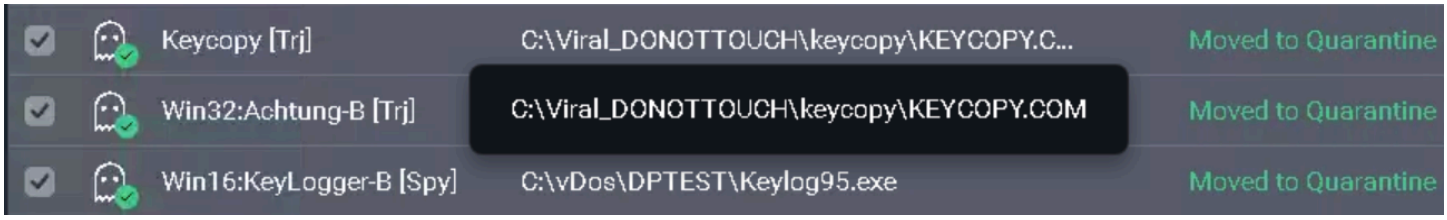
Sample Incident Report: http://www.vlabsondemand.com/issa30/SecurityIncidentReport-Sample.pdf

Part 2: Use Antivirus Software to Scan the Infected System



Screenshot 1: Summary of the deep scan performed on TargetWindows02 with AVG after connecting to the remote admin server

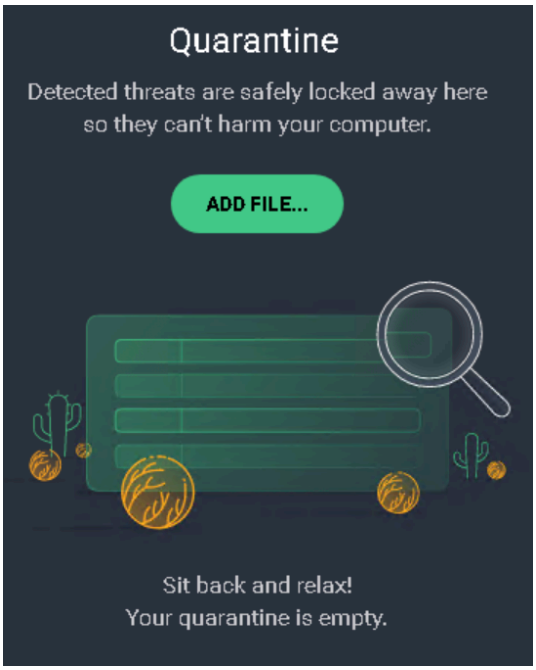
Part 3: Investigate and Remove the Identified Threats



Screenshot 2: Full file path of KEYCOPY.com threat

Document any information you discover about eradicating this threat:

Keycopy.com is a type of malware that tracks user activity, such as keystrokes, applications used, and history. This spyware can then gather sensitive information, such as passwords, credit card numbers, and PIN numbers, and send it to a third party without the device owner's consent. The first step is to prevent the potential spread of this malware by disconnecting the device from the internet. Use antivirus or antimalware to scan the entire device thoroughly and identify any files associated with keycopy.com to quarantine and remove them. On Windows, clean the Windows Registry by removing invalid registry entries using a registry cleaner program. It is also a good idea to check browser extensions and plugins. Check all web browsers for any suspicious extensions or plugins, remove them, clear browser cache and cookies, and reset browser settings. This can remove any unwanted changes made by the malware. As always, ensure that the operating system and all software are up to date with the latest security patches. Additionally, change any passwords or other credentials for important accounts that may have been compromised immediately. Continue to monitor the device for any continued suspicious activity to potentially help detect and respond to any lingering issues or potential new threats.



Screenshot 3: All infected files have been permanently deleted, leaving quarantine empty.

Security Incident Report

Incident Report #: J&B-Lab07

Reported Date and Time: April 18, 2024

Technician: Gianna Julio

Site Location: TargetWindows02 (172.30.0.10)

Identification (Type and how detected): Routine deep scan of TargetWindows02 with Anti-Virus Guard (AVG)

Virus scan detected: keycopy.com and achtung.exe

Triage (Impact): Malware only affected the single device and did not spread within the network.

Containment (Steps taken):

1. Ran a manual virus deep scan which identified the malware and placed it in quarantine.
2. Permanently removed all affected files in quarantine from the system.

Investigation (Cause): File shield and web shield features of basic AVG protection were turned off.

Recovery and Repair (Resolution):

- Used Antivirus software to quarantine and eradicate the malware.
- Implemented scanning of corporate email for malware and spam.

Lessons Learned (Debriefing and Feedback):

- Antivirus software on systems should be configured to scan all hard drives regularly to identify any file system infections and, optionally, to scan other storage media as well.
- Users should be able to launch a scan manually as needed.
- Users should be educated on protecting themselves from viruses by running only company-authorized Antivirus software, not opening suspicious email attachments, and not responding to suspicious emails.